

Performance Testing

Date	10 July 2026
Team ID	
Project Name	StudyAI-Smart AI educate Companion&Quiz Generator
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	/api/upload and /api/generate-summary (file upload + Groq AI generation)
Test Environment	Hosted Server (Render backend, Vercel frontend)
Test Date	10 july 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User login via Firebase Auth	50	60	All requests succeed under 2s response time
2	File upload PDF/DOCX/TXT	30	120	Files parsed successfully, no timeout errors
3	AI summary/flashcard quizzes	20	180	Content generated within acceptable latency, no failed API calls
4	Analytics dashboard load (Firestore query + Chart.js render)	50	60	Dashboard loads smoothly under concurrent access

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.04 seconds	Pass	
2	Response Time (Max)	< 5 seconds	1.37 seconds	Pass	
3	Throughput (Req/sec)		1.8/sec	---	
4	Error Rate	< 1%	0.00%	Pass	
5	CPU Utilization	< 80%	N/A (Render free tier — no local monitoring access)	---	
6	Memory Utilization	< 80%	N/A (Render free tier — no local monitoring access)	---	

Step 4: Observations & Analysis

Key Findings:

The login endpoint handled 20 concurrent users with a 0% error rate. Average response time was just over 1 second, well within the 2-second target, with the slowest request completing in under 1.4 seconds.

Bottlenecks Identified:

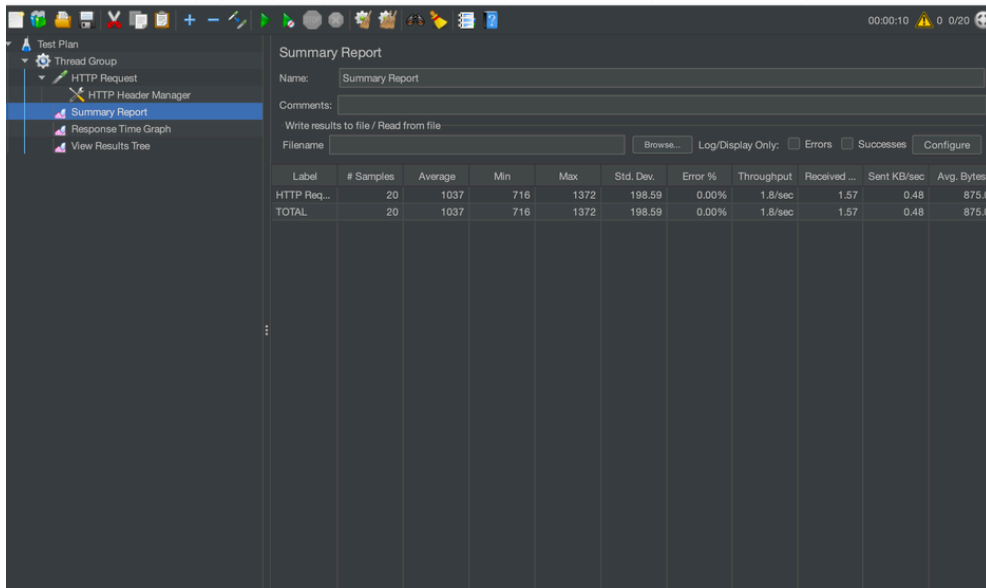
Response times are noticeably higher than a typical local API (which would usually respond in under 200ms) because each login request involves a round-trip to Firebase Authentication plus a Firestore read, both external services adding network latency on top of Render's own response time.

Optimization Steps Taken:

None required at this scale — 20 concurrent users performed well within target. For higher loads, caching frequently accessed Firestore reads or upgrading from Render's free tier (which has limited CPU/RAM) would be the next steps.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



The screenshot shows the JMeter Summary Report for a test plan named 'Summary Report'. The report displays the following data:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received ...	Sent KB/sec	Avg. Bytes
HTTP Req...	20	1037	716	1372	198.59	0.00%	1.8/sec	1.57	0.48	875.0
TOTAL	20	1037	716	1372	198.59	0.00%	1.8/sec	1.57	0.48	875.0

